FACULTY OF ECONOMIC AND MANAGEMENT SCIENCES

DEPARTMENT OF ECONOMICS

ECONOMICS 114 PRACTICE TEST

Total Marks:	40
Examiners:	Mr J.H. Pretorius

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.
ANSWER SECTION A's QUESTIONS ON THE SUPPLIED ANSWER SHEET. INDICATE YOUR CHOICE BY COLOURING THE CIRCLE NEXT TO THE NUMBER OF THE STATEMENT.
ANSWER SECTION B's QUESTIONS IN THE ANSWER BOOK PROVIDED. START EVERY QUESTION ON A NEW PAGE. SHOW ALL CALCULATIONS.

Key Abbreviation	**Meaning**	**Key Abbreviation**	**Meaning**
AD	Aggregate demand	MRT	Marginal rate of transformation
APL	Average product of labour	N	Employment
C	Consumption expenditure	NX	Net exports
CPI	Consumer price index	P	Price level
FF	Feasible frontier	PS	Price-setting
G	Government expenditure	SARB	South African Reserve Bank
GDP	Gross domestic product	TP	Total product
I	Investment	U	Unemployment rate
M	Import expenditure	W	Nominal wage
MP	Marginal product	w	Real wage
MPC	Marginal propensity to consume	WS	Wage-setting
MPS	Marginal propensity to save	X	Export expenditure
MRS	Marginal rate of substitution	Y	Current income

SECTION A

TRUE OR FALSE STATEMENTS

[4 marks]

(1 mark for a correct answer and $-\frac{1}{2}$ for an incorrect answer)

1. If a De Lapa robot costs R100 and two minute noodles cost R20, then the relative price of a robot is 0,2 in terms of noodles.

   a) True

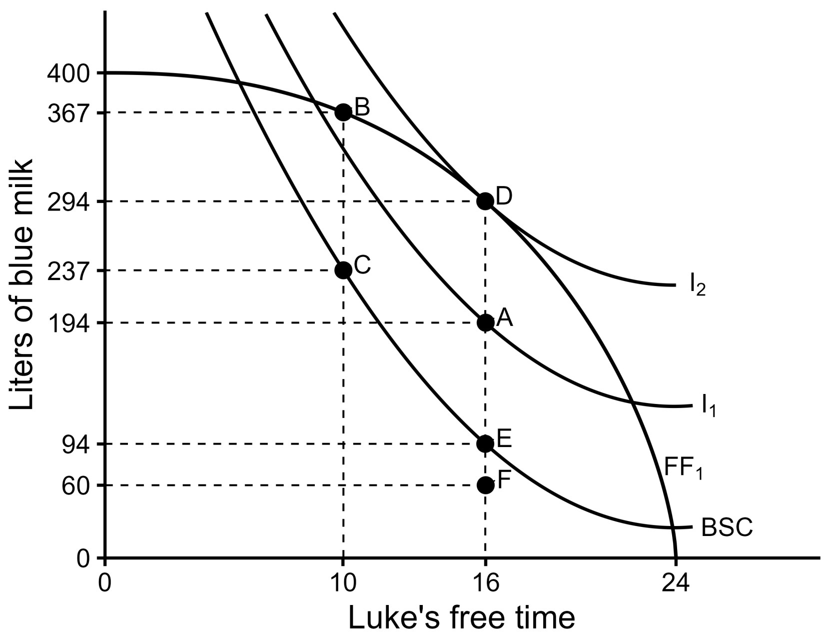
   b) False

2. Over the past century, average working hours worldwide have declined, as the substitution effect of rising wages outweighed the income effect.

   a) True

   b) False

Consider the figure below to answer questions 3 and 4. The figure illustrates the combined feasible frontier for Luke, a blue milk farmer, and Owen, the owner of the farm. Assume that Owen forces Luke to work on the farm.



3. Point C illustrates an allocation where Luke works 14 hours per day and produces 367 liters and gets 237 liters, while Owen gets 130 liters.

   a) True

   b) False

4. Allocation C is not an optimal allocation, because $MRS > MRT$ and Luke will be better off if Owen gives him more free time.

   a) True

   b) False

SECTION B

MULTIPLE CHOICE QUESTIONS

[10 marks]

(2 marks for a correct answer and $-\frac{1}{4}$ for an incorrect answer)

5. Epic Island's GDP was 200 V-bucks (their currency) in 2024. If the annual GDP growth rate was $2,3\%$, what was Epic's GDP in 2025?

   a) 204,0
   b) 204,7
   c) 195,4
   d) 204,6

Aqueel and Curtly both like to speed, and like racing on their time off. When they both speed, they have the potential of causing a motor accident and injuring innocent bystanders. When only one speeds, he will win the race and the one that doesn't, will get caught by the traffic officer and be penalized. Answer Questions 15 and 16

		Aqueel	
		Don't Speed	**Speed**
Curtly	**Don't Speed**	4, 4	1, 5
	Speed	5, 1	2, 2

6. The Nash Equilibrium of the game shown above is:

   a) Don't speed, Don't speed
   b) Don't speed, Speed
   c) Speed, Don't speed
   d) Speed, Speed

7. This game is an example of what type of game?

   a) ultimatum game
   b) invisible hand game
   c) prisoner's dilemma
   d) chicken game

8. Dorp and Punk are two neighbouring pubs in Stellenbosch. They serve two beverages, beer and brandy, each night from 6pm to 12am. Each pub therefore has 6 hours per night available to produce these drinks. If Punk produces both beverages themselves, they will spend 3 hours pouring 50 beers and 3 hours pouring 33.5 brandies. The table below shows how much beer and brandy Dorp and Punk would be able to produce if they spent 100% of their time producing only one beverage:

	Production if 100% of time spent	
	Beer	**Brandy**
Dorp	120	95
Punk	x	67

Which one of the following statements are TRUE?

a) Punk has an absolute advantage pouring beers

b) Dorp has a comparative advantage pouring brandies

c) There are no gains from trade should Punk and Dorp choose to trade

d) Dorp has a comparative advantage pouring beers

9. A student receives R100 pocket money each week. They have to decide how much money to spend on food and socialising every week. Assume food and socialising are ‘normal goods’. If the price of food decreases, *ceteris paribus*, socialising will experience a:

a) negative income effect, negative substition effect

b) positive income effect, positive substition effect

c) negative income effect, positive substition effect

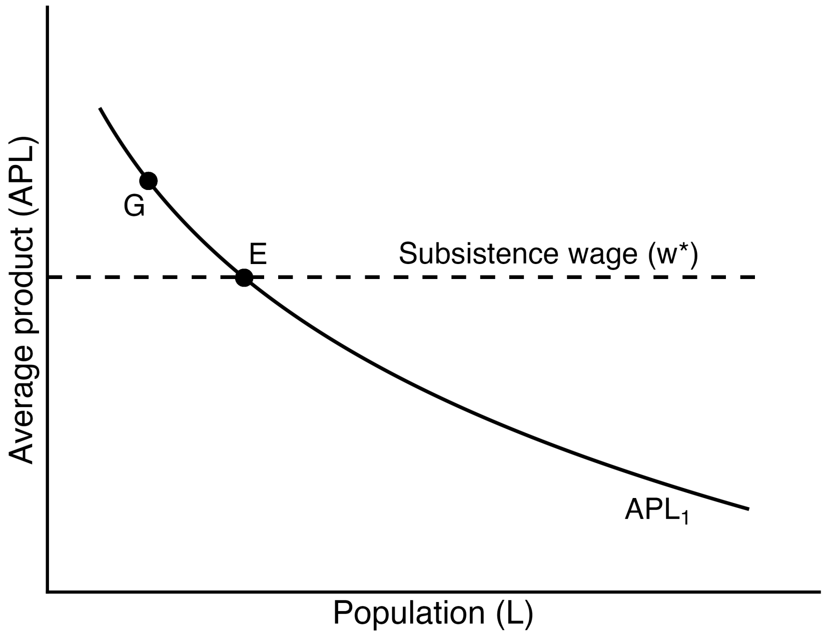
d) positive income effect, negative substition effect

SECTION C

DESCRIPTIVE QUESTIONS
[20 marks]

Question 10 [6 marks]

Refer to the graph provided below. A Malthusian economy is currently at point G, where the Average Product of Labour (APL) falls above the subsistence level (w*). Assume that population growth responds gradually to changes in income and that land and labour are the only inputs in production ceteris paribus.



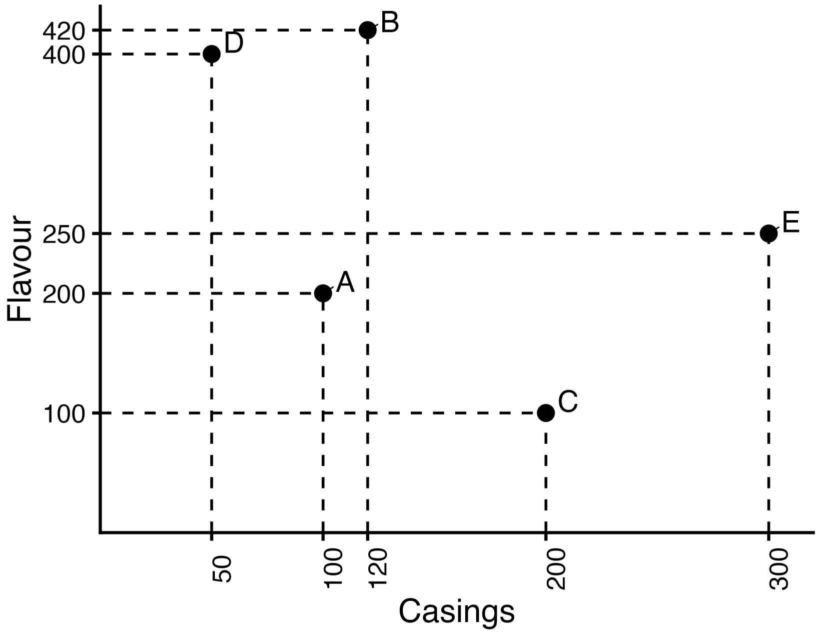
10.1 Is this economy in long-run equilibrium? Explain. [2]

10.2 Why does the APL curve become flatter at higher populations?. [2]

10.3 Explain what will happen to the economy over the long-run if it is at point G. [2]

Question 11 [4 marks]

Consider the graph below, which shows five different technologies (A, B, C, D, and E) for producing 500 Zesty Lemon vapes per year in a factory, using different amounts of casings and flavour. We also know that the factory's total revenue per year from the sale of vapes is R55 000 [We assume that the factory size and the total output remain the same throughout this exercise: *ceteris paribus*]

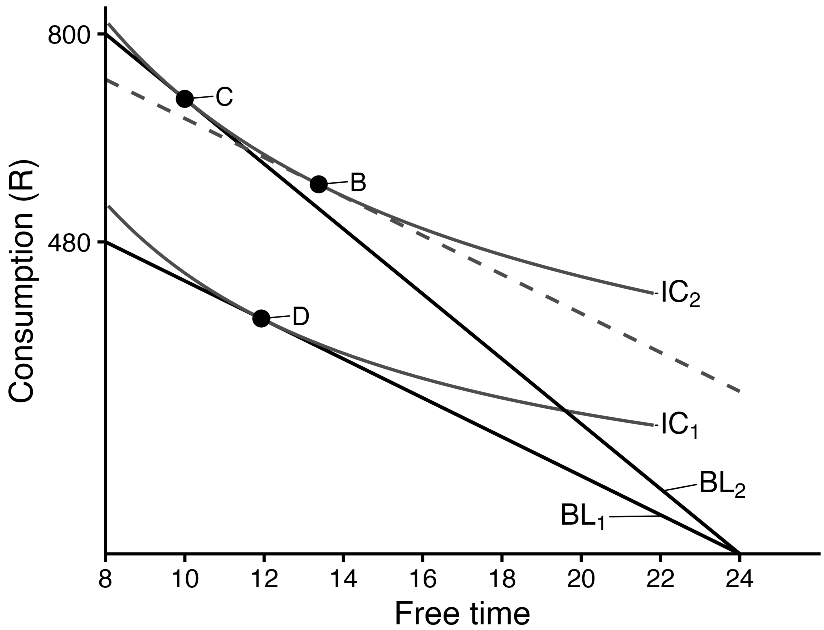


11.1 Without knowing the cost of casings or flavour, which technologies will the factory not choose? Explain. [2]

11.2 If casings and flavour cost R20 and R50, respectively, derive the equation for the isocost curve of technology A. [2]

Question 12 [7 marks]

The graph below shows two feasible sets (BL_1 and BL_2), as well as two indifference curves for Chad, an employee at Aura Farm. Due to his commitment to mogging, Aura Farm gives Chad an hourly wage increase. Answer the questions below.



12.1 By how much did Chad's wage increase? [2]

12.2 What is the MRS of the optimal consumption bundle after the wage increase? [1]

12.3 The movement between which points on the figure indicate the income and substitution effects? [2]

12.4 With respect to free time, does the income or substitution effect dominate? Explain. [2]

—— **END** ——